Fig. 3: Home page of the Metaverse Standards Forum

tunities to rethink user interaction, distribution, and revenue channels, as well as how (and where) work is orchestrated and conducted.

There are several neighbouring yet distinct technologies that will need to be integrated and adopted into the metaverse for this to be fully realised. Amongst the most important of these are blockchain technologies, cloud infrastructure, Internet of Things (IoT), artificial intelligence, machine learning, robotics, and connectivity-based infrastructure such as satellite internet constellations and 5G/6G—all built on or predominantly open source.

Open source holds a critical role in the development of metaverse-aligned technologies and establishing the standards that will underpin the technology. There are numerous macro challenges that present significant obstacles to wider adoption of the metaverse—for example, interoperability, identity, privacy, security, sustainability, governance, and inclusivity. Open source software, projects, and communities are essential for addressing and agreeing upon the best way to satisfy these challenges.

Blockchain technology will play an extremely vital role in surmounting these challenges and, consequently, many of the associated projects and communities of practice are open source. A prime example of this surrounds the projects focused on self-sovereign identity (SSI), which is an approach that gives individuals control of their digital identities and addresses the difficulty of establishing trust in an interaction. Given the transient and virtual/remote nature of human interaction in the metaverse, this question of identity, ownership, and trust is at the heart of whether the metaverse is a safe, inclusive, and mutually beneficial place to work, socialise, and play, or the very opposite.

The metaverse is increasingly becoming a viable place to work and to sell products and services, and where organisations should focus their attention, effort, and resources. The metaverse market is still at the foot of its growth curve and the associated technologies are still in the formative stages. As such, some of the key building blocks are still works in progress.

Much of this work is being conducted in open source communities and projects, and represents a fabulous opportunity for open source oriented organisations to 'get in on the ground floor' and cement their position as thought leaders and pioneers in this exciting and emerging technology.

Metaverse components and open source

Now let us consider some of the key components that will facilitate the ability and scalability of working in the metaverse.

Identity. The metaverse and its future are intrinsically linked to web3, which is billed as the next iteration of the World Wide Web. It is based on blockchain technology, which incorporates concepts such as decentralisation and token-based economics. The key term here is 'decentralised,' i.e., not managed or controlled by a centralised body, or company.

In the present iteration, web2, the market and our data (including our digital identity) is monopolised and monetised by a handful of huge technology companies such as Meta (previously Facebook), Google, Microsoft, etc.

A key tenet of web3 is to liberate, or democratise and decentralise our personal data, including our digital identity. There will, at least in first phase, not be one single, vast metaverse, but thousands of metaverses.

In the future, some predict that, as there is one internet, there will be one metaverse, interconnecting multiple sub-metaverses in one coherent, standards-based experience. Some metaverses will be private, some